

Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

Top 3 Risk Factors :

- **High Credit Utilization (>75%)** – Indicates financial stress and over-reliance on credit.
- **Multiple Missed Payments (2 or more)** – Strong historical signal of future delinquency.
- **Low Credit Score (<600)** – Reflects poor repayment behavior and higher default risk.

Key Insight	Customer Segment	Influencing Variables	Potential Impact
High delinquency risk due to overutilization and missed payments	Customers under 30 with 2+ missed payments	Credit_Utilization, Missed_Payments, Age, Debt_to_Income_Ratio	Enables targeted outreach to reduce delinquency and improve recovery

2. Recommendation Framework

Restated Insight:

Customers with credit utilization above 75% and 2 or more missed payments are at significantly higher risk of delinquency.

SMART Recommendation:

“Launch a targeted 6-week SMS and email outreach campaign focused on high-risk customers, aiming to reduce 30+ day delinquency by 12%.”

Justification:

This recommendation is highly focused, actionable using current tools, and addresses the highest-risk segment identified by the model. It is cost-effective, quick to deploy, and aligns directly with Geldium's goals to reduce default rates while maintaining customer trust and fairness.

3. Ethical and Responsible AI Considerations

Fairness Risk 1:

The model might overpredict risk for low-income or unemployed customers.

Mitigation: Monitor prediction patterns across segments and adjust with fairness constraints.

Fairness Risk 2:

Certain features like location or employment status could act as proxies for protected attributes.

Mitigation: Conduct bias audits (e.g., disparate impact analysis) and exclude sensitive features if needed.

Explainability:

Using Logistic Regression ensures that every prediction can be linked back to clear variables (e.g., credit utilization, payment history), making it understandable to both business users and customers.

Responsible AI:

All recommendations are supportive and not punitive. Outreach efforts aim to inform and assist customers, reflecting Geldium's values and commitment to fairness and ethical decision-making.